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Using Scientific Notation

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Using Scientific Notation

Scientific notation is a system for writing very large or very small numbers. This system is based on the powers of ten. The chart below shows some of the powers of ten. Do you notice any patterns in the numbers?

$10^5 = 100,000$	$10^2 = 100$
$10^4 = 10,000$	$10^1 = 10$
$10^3 = 1,000$	$10^0 = 1$

In each case, the exponent tells you how many zeros are in the number or how many places come after the 1.

A number in scientific notation includes a number part and a power of ten. The number part has a digit in the ones place followed by a decimal point and any other important digits. The exponent is the number of places the decimal point had to move.

Example 1

Hide

Write 2,800,000 in scientific notation.

Step 1

Move the decimal point from the end of the number to the place directly after the first non-zero digit. Write the number part as 2.8.

Step 2

Count the number of places you moved the decimal place to the left.

2.800,000.
← 6 places The decimal point moved left 6 places. Use 6 for the exponent.

Write the power of ten as 10^6 .
2,800,000 is written 2.8×10^6 .

Example 2

Hide

Write 0.000056 in scientific notation.

Step 1

Move the decimal point from the start of the number to one place after the first non-zero digit. Write the number part as 5.6.

Step 2

Count the number of places you moved the decimal place to the right.

0.00005.6
→ 5 places The decimal point moved right 5 places. Use -5 for the exponent.

Write the power of ten as 10^{-5} .
Note that smaller values are written in scientific notation using a negative exponent.
0.000056 is written 5.6×10^{-5} .

To change a number from scientific notation, work backward. Move the decimal point to the right for a positive exponent and to the left for a negative exponent. Add zeros for placeholders as needed.

Remember, if the exponent is positive, the actual number is much larger. If it is negative, the actual number is much smaller.

Example 3

Hide

3.14×10^7 Move the decimal point 7 places to the right.
3.1400000.
→ 7 places $3.14 \times 10^7 = 31,400,000$

2.6×10^{-4} Move the decimal point 4 places to the left.
.0002.6
← 4 places $2.6 \times 10^{-4} = 0.00026$

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Category

Lesson 1: Exponents and Roots

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